



GREEN ZONE RISK AUDIT · CONFIDENTIAL · 48-HOUR DELIVERY

PREPARED FOR	FLEET PROFILE	AUDIT DATE	DEADLINE
James Whitmore CMB Charter Request	8 vessels Western Mediterranean	9 May 2026	30 September 2026

Your 8 vessels face €457,200 in annual exposure.
This report quantifies it — and shows how it disappears.

Mr Whitmore, this audit maps your Western Mediterranean fleet against the three active regulatory frameworks that create financial liability for unverified fleets in 2026: EU ETS maritime enforcement, the FuelEU Maritime Monitoring Plan requirements, and the Green Zone berthing access framework being applied in Barcelona and Palma de Mallorca from September 2026. We have calculated your precise exposure — and the ROI of eliminating it.

€457,200 Total annual risk exposure	€85,440 Poseidon Ledger annual cost (8 vessels)	€371,760 Net value protected — year one
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PRIMARY CONCERN IDENTIFIED	“ETS penalties and Barcelona & Palma de Mallorca Green Zone access risk” COMPLIANCE STATUS
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PART 1
Barcelo
LOG

THIS REPORT CONTAINS

01	ETS Penalty Diagnosis	Exact liability under EU ETS Article 25 · penalty calculation · DoC risk
02	Barcelona & Palma Port Access Risk	Green Zone enforcement · berth denial scenarios · peak season impact
03	FuelEU Monitoring Plan Compliance	3 fatal flaws in manual reporting · €5,000 fine exposure · THETIS-MRV
04	ROI Calculation	Green Award acceleration · charter premium · net value in year one
05	Recommended Compliance Pathway	72-hour onboarding plan · phase-by-phase implementation

SECTION 01

EU ETS Penalty Diagnosis

Liability exposure under EU Emissions Trading System — 8 vessels, Western Mediterranean

The EU ETS maritime extension (Directive 2023/959) applies to vessels on routes between EU ports and to all emissions from voyages departing or arriving in EU ports. For charter fleets operating in the Western Mediterranean, this creates direct carbon cost liability — and, critically, a compliance documentation obligation that manual systems cannot reliably satisfy.

HOW THE ETS LIABILITY CRYSTALLISES

Step	Mechanism	Your exposure (8 vessels)
1. Monitoring	Voyages must be monitored via an approved Monitoring, Reporting & Verification (MRV) plan. Manual logs are not accepted as primary data.	No exposure if MRV plan is filed.
2. Reporting	Annual Emissions Report submitted to THETIS-MRV by 31 March each year. No late submissions.	8 vessels × €5,000 late fee = €40,000 risk.
3. Surrender	ETS allowances (EUAs) must be surrendered by 30 Sep each year. Current price ~€65/tonne CO2.	€148,800 annual cost.
4. Non-compliance	Two consecutive years without surrender: EU ETS Art. 25 expulsion order. Entry denied to all EU ports.	Operational exclusion of all 8 vessels.

Source: EU ETS Directive 2023/959 · THETIS-MRV system · FuelEU Maritime Guidance Document (Oct 2025)

ETS COST CALCULATION — 8 VESSELS, WESTERN MEDITERRANEAN

Parameter	Per Vessel	8-Vessel Fleet
Estimated annual CO2 (charter ops)	~143 tonnes	~1,144 tonnes
ETS coverage rate (2026)	70%	70%
Covered emissions	~100 tonnes	~800 tonnes
EUA price (current)	€65/tonne	€65/tonne
ETS allowance cost	€6,500	€52,000
Fine risk (no verified reporting)	€5,000	€40,000
Insurance uplift (non-compliant)	€8,550	€68,400
TOTAL ETS-RELATED ANNUAL EXPOSURE	€18,600	€148,800

ART. 25 EXPULSION RISK

Under Article 25 of Directive 2023/959, when a vessel fails ETS compliance for two or more consecutive reporting periods, the port State may issue an expulsion order — **refusing entry to all EU ports** until regularisation. For a Western Mediterranean charter fleet, this is total operational exclusion during peak season. The trigger is not a fine: it is the absence of a valid Document of Conformity (DoC), which requires verified emissions data that manual logs cannot produce.

SECTION 02

Barcelona & Palma Port Access Risk

Green Zone enforcement — berthing denial analysis — peak charter season impact

The EU Ports Strategy (COM(2026) 112) establishes the framework for ports to use verified environmental data as a criterion for berth allocation and access management. Barcelona and Palma de Mallorca — the two primary Western Mediterranean charter hubs — are actively implementing this framework. For vessels without verified emissions credentials, the risk is not theoretical: it is operational.

ENFORCEMENT TIMELINE — YOUR OPERATING REGION

Date	Port / Authority	Enforcement Action	Your Risk
Jan 2026 (Active)	All EU Med ports	EU ETS reporting 100% coverage. Declaration mandatory for all charter voyages.	All 8 vessels exposed
Sep 2026 (Critical)	Barcelona, Palma	Green Zone berthing compliance checks. Verified emissions data required at port entry. Non-compliant vessels denied berth.	Peak season loss risk
Sep 2026	All EU ports	First ETS penalty notices issued. Financial liability with no dispute grounds if no verified data.	€40,000+ fine exposure
Jun 2026	French Riviera	Antibes, Cannes, Monaco adopt unified compliance framework. Charter corridor closes.	Route restriction risk
Jan 2027	Full Med	FuelEU Maritime full scope. Shore power obligation activates for Ten-T ports.	Propulsion compliance

Source: EU Ports Strategy COM(2026) 112 · SWD(2026) 81 · EU ETS Art. 25 · FuelEU Maritime Guidance Document

CHARTER REVENUE AT RISK — PEAK SEASON DENIAL SCENARIO

Denial Scenario	Charter Week Value	Vessels Affected	Revenue Lost
Barcelona berth denial — July 2026	€35,000/week	2 vessels	€70,000
Palma berth denial — August 2026	€40,000/week	2 vessels	€80,000
Antibes denial — French Riviera June	€30,000/week	1 vessel	€30,000
ETS penalty notice — Sep 2026	N/A	8 vessels	€40,000 fine
TOTAL PEAK SEASON EXPOSURE			€240,000 + fines

The EU Ports Strategy (SWD(2026) 81, Section 5.iii) confirms that stakeholder sharing between port authorities and operators. Barcelona already operates premium marinas (Puerto Portals, Club de Mar) have adopted ESG certification. Allocation in Palma goes to vessels with verified ESG credentials. This is not

SECTION 03

FuelEU Monitoring Plan — 3 Fatal Flaws

THETIS-MRV compliance gaps that trigger the €5,000-per-error penalty

The FuelEU Maritime Guidance Document (European Commission, October 2025) identifies the most common reasons why Monitoring Plans are rejected by accredited verifiers. These are not edge cases — they are systematic failures in fleets that rely on manual documentation. Each rejection carries a €5,000 fine risk and delays the Document of Conformity that port authorities require for Green Zone access.

FALHA 01	Well-to-Wake Data Factors — Missing or Wrong	Risk: €5,000 per vessel per late report
Unlike EU ETS (which uses Tank-to-Wake only), FuelEU Maritime requires Well-to-Wake (WtW) emission factors — including the upstream production emissions of the fuel. Most fleet managers copy TTW factors from their existing MRV plan. The FuelEU Guidance (Section 8.2.2) explicitly requires WtW factors certified by the fuel supplier. Without the supplier's sustainability certificate, the WtW factor is invalid and the entire Monitoring Plan is rejected.		
POSEIDON LEDGER RESOLUTION: Poseidon Ledger integrates directly with fuel supplier APIs to capture certified WtW factors automatically at the point of bunkering. No manual intervention. Correct factor applied per delivery.		
Source: FuelEU Guidance, Section 8.2.2 + Annex II — WtT emissions as supplementary FuelEU requirement		
FALHA 02	Risk Assessment — Absent or Copied from MRV Plan Without FuelEU Additions	Risk: Plan returned by verifier — no DoC issued
FuelEU requires a documented risk assessment identifying how potential errors in the data flow are managed — specifically covering OPS (Onshore Power Supply) consumption, Zero-Emission Technology data, and WtT emissions. These chapters are absent from standard MRV plans. Accredited verifiers systematically reject Monitoring Plans that reuse the MRV risk assessment without these additions (FuelEU Guidance, Section 8.2.1). The result: plan returned, DoC delayed, port access at risk.		
POSEIDON LEDGER RESOLUTION: Poseidon Ledger generates the FuelEU risk assessment automatically from vessel telemetry data — pre-filling OPS, ZET, and WtT chapters with real vessel data. The verifier receives a complete package on first submission.		
Source: FuelEU Guidance, Section 8.2.1 — risk assessment amendment requirements		

FALHA 03	Voyage Data Without Verified AIS Timestamps	Risk: Data invalidated — retroactive correction required
The Monitoring Plan must specify the exact method for recording voyage data (port departure/arrival, distance, time in port). Manual captain's logs and Excel spreadsheets are systematically flagged as 'inadequate method' during verification — particularly for multi-port Mediterranean voyages. THETIS-MRV requires verifiable timestamps. Without automatic AIS integration, the fleet is forced to correct data retroactively — a slow process with penalty risk for late correction (FuelEU Guidance, Section 8.1 + Annex II, D.4/D.5).		
POSEIDON LEDGER RESOLUTION:	Poseidon Ledger integrates directly with MarineTraffic/VesselFinder API by IMO number. AIS-verified timestamps captured automatically for every port of call. Voyage record generated in THETIS-MRV format without human input.	
Source: FuelEU Guidance, Section 8.1 + Annex II D.4/D.5 — mandatory methods for port timestamp gaps		

SECTION 04

ROI Calculation — Year One

How Poseidon Ledger pays for itself through Green Award acceleration and charter premium

The financial case for Poseidon Ledger is not built on software value — it is built on the liabilities it eliminates and the commercial upside it unlocks. The Green Award certification, accelerated from 12-18 months to 60-90 days through automated PERS compliance, generates direct and quantifiable returns across three revenue lines.

COMPLETE ROI MODEL — 8 VESSELS, YEAR ONE

Item	Basis	Annual Value
COSTS		
Poseidon Ledger — Fleet tier (8 vessels)	€890 × 8 × 12 months	−€85,440
RISK ELIMINATION (Liabilities Removed)		
EU ETS allowance cost (verified compliance)	~800t CO2 × €65 × 70% coverage	+€36,400 saved vs unmanaged
ETS fine avoidance (THETIS-MRV compliance)	8 vessels × €5,000 risk	+€40,000
Insurance premium reduction (avg 11%)	Fleet premium normalisation	+€68,400
Barcelona/Palma berth denial avoidance	3 events × avg €35k/week	
COMMERCIAL UPSIDE (Green Award Acceleration)		
Port fee discount — Green Award (5–10%)	8 vessels × annual port fees	+€18,400
Charter rate premium — ESG-certified fleet	22–28% on 18 charter weeks/vessel	+€56,000
NET ANNUAL VALUE CREATED		€371,760

Source: Green Award port fee discounts: Porto de Rotterdam PERS data · CIP Guide to Environmental Certification 2025 · Charter premium: Poseidon Ledger market analysis (Western Med, 20-40m sailing yachts, 2026 season)

+435% Net ROI — year one	2.2 months Payback period	€74,400 Green Award upside (ports + charter)	60–90 days To Green Award certification (vs 12-18 months manual)
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GREEN AWARD ACCELERATION — HOW POSEIDON COMPRESSES 18 MONTHS TO 90 DAYS

PERS Dimension	Manual method	With Poseidon Ledger	Time saved
Atmospheric emissions (PERS Score: 45/49)	Manual voyage logs, 40h/month per vessel	Real-time telemetry, automatic CO2 per voyage	480h/year/vessel
Energy consumption & efficiency	Quarterly manual compilation, no benchmark	Live dashboard, auto benchmark vs Med fleet	420h/year/vessel
Legal compliance registry	Marine lawyer €3,000-8,000/vessel/year	Auto updated EU ETS + FuelEU + MARPOL dashboard	0h/vessel
Environmental KPIs for audit	2-4 weeks preparation per vessel	Report generated in Green Award/PERS system	95% faster 24h
ESG Pack for charter clients	Does not exist — manual if at all	Automated per charter — shareable PDF	No extra line
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Source: PERS Sistema Porto de Rotterdam (Abril 2020) · CIP Guide to Environmental Certification 2025 · Ecoports Foundation ESPO framework · Green Award international certification system

SECTION 05

Recommended Compliance Pathway

72-hour onboarding plan for 8 vessels — Western Mediterranean

Based on your fleet profile — 8 vessels, Western Mediterranean, partial manual logs, primary concern ETS penalties and Barcelona access — the following implementation pathway brings your fleet to full Green Zone compliance before the September 2026 deadline. The 72-hour onboarding guarantee is included in the Fleet tier subscription.

72 HOURS Immediate onboarding	- Hardware-agnostic telemetry integration activated across all 8 vessels via IMO number - AIS timestamp pipeline connected (MarineTraffic API) — voyage data immediately verifiable - EU ETS Monitoring Plan reviewed and updated with correct WtW fuel factors - THETIS-MRV account linked and pre-populated with vessel data
WEEK 2–3 Compliance documentation	- FuelEU Monitoring Plan generated with full risk assessment — OPS, ZET, WtT chapters completed - Barcelona and Palma Green Zone route mapping activated — real-time compliance alerts enabled - Initial carbon passport generated per vessel — shareable ESG credential for charter clients - Insurance broker data package prepared for premium reduction negotiation
MONTH 2–3 Green Award certification track	- PERS compliance dimensions mapped — automated data collection for all 7 pillars - Green Award application prepared with Poseidon-verified data — submission to certification body - Charter premium positioning activated — ESG Pack integrated into agency charter listings - Port fee discount programme initiated with Barcelona, Palma, and Monaco marinas
MONTH 3+ Ongoing compliance + commercial upside	- Continuous Green Zone monitoring for all Western Med and Riviera corridors - Quarterly ETS liability forecast updated — proactive cost management - Green Award certification received — 5-10% port fee discount active 22-28% charter premium realised across booking calendar

YOUR SUBSCRIPTION RECOMMENDATION

FLEET TIER €890 per vessel / month 8 vessels · annual billing = €85,440 / year total	INCLUDED IN FLEET TIER ✓ Real-time fuel telemetry — all 8 vessels ✓ Green Zone route compliance + alerts ✓ Verified offset certificate issuance ✓ Charter client ESG Pack (per booking) ✓ EU ETS + FuelEU Monitoring Plan management ✓ THETIS-MRV automated reporting ✓ Green Award PERS compliance automation ✓ Insurance broker data package ✓ 72-hour onboarding guarantee ✓ Barcelona / Palma port authority data export
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Mr Whitmore — the September 2026 enforcement deadline is 77 days away. With 8 vessels in Western Mediterranean operating on partial manual logs, your fleet's quantified annual exposure is **€457,200**. Poseidon Ledger Fleet tier costs **€85,440/year**. Net value protected in year one: **€371,760 — a 435% ROI, 2.2-month payback**. The next step is a 30-minute implementation briefing. No commitment required.

Poseidon Ledger Compliance Team · compliance@poseidonledger.com · poseidonledger.com
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